

PARAMETRIC HYDRODYNAMIC ANALYSIS OF TWO-BODY POINT ABSORBER WAVE ENERGY CONVERTERS

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ABSTRACT

This work presents a comprehensive parametric analysis of the hydrodynamic response of two-body Point Absorber (PA) Wave Energy Converters (WECs), focusing on the influence of floater geometry on system dynamics as a preliminary phase toward optimization design. The hydrodynamic coefficients—added mass (A), radiation damping (B), and excitation force (F_e)—are computed using CAPYTAIN, and mean absorbed power is estimated under two realistic sea state conditions while accounting for operational constraints on relative displacements. The results are analyzed considering the variation of two geometric parameters, the R/D ratio, defined as the ratio between the floater radius R and the external draft D , and D_{int} , defined as the internal draft. Floaters with lower R/D values are found unsuitable as they exceed displacement limits, whereas intermediate ratios enhance power capture by tuning the natural frequency of relative heave motion to the predominant wave frequencies, thereby promoting resonance and amplifying response. The results also show that variations in R/D lead to significant changes in mean absorbed power, while changes in D_{int} have only minor effects.

Keywords: Wave Energy Converter (WEC), Point Absorber (PA), Hydrodynamic Analysis, Boundary Element Method (BEM), Real Sea States

1. INTRODUCTION

Marine Renewable Energy (MRE) has emerged as a promising contributor to the decarbonization of the power sector by harnessing the vast energy contained in ocean waves, tides, and currents. In the United States, the technical potential of these resources is estimated to satisfy up to 50% of the national electricity demand [1, 2]. Although challenges persist in integrating MRE systems into the main power grid—due to their site-specific

and intermittent nature—the Powering Blue Economy (PBE) applications, such as sustainable aquaculture, offshore charging stations for autonomous underwater vehicles and coastal resilience infrastructure, offer a viable near-term pathway for practical deployment of MRE technologies [3].

Wave Energy Converters (WECs) represent a key class of technologies to extract usable power from ocean wave motion, encompassing diverse operating principles, geometries, and control strategies [4]. In the United States (U.S.), WEC development is increasingly aligned with PBE priorities [3]. Among WEC concepts, point absorbers (PAs) have attracted considerable attention due to their compact footprint, omni-directional wave capture, and favorable power-to-size ratio [5]. Single- and two-body configurations are actively investigated to exploit relative heave motion, thereby enhancing energy absorption and providing greater flexibility in tuning and control. Recent studies have explored experimental modeling and validation of PA arrays [6], the incorporation of tuned inerters in two-body designs to improve power capture through enhanced inertial effects [7], and the dynamic coupling between floating PAs and catenary mooring systems [8]. However, despite these advances, the systematic influence of floater geometry on hydrodynamic coefficients, system dynamics, resonance behavior, and energy conversion in two-body PAs remains underexplored in a comprehensive, parametric manner. This gap limits the development of optimized, geometry-informed designs tailored to near-term PBE applications.

Given this identified need for a more systematic examination of geometry-dependent effects in two-body PAs, the Reference Model 3 (RM3) emerges as a particularly valuable and widely recognized benchmark [9]. Developed under Reference Model Project (RMP) of the U.S. Department of Energy (DoE), the RM3 consists of a floating buoy oscillating relative to a submerged reaction structure. Originally scaled for grid-connected applications, it has undergone extensive numerical and experimental validation, resulting in a robust, well-documented dataset

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that is extensively used by the WEC research community, undergone extensive wave tank testing [10] to validate numerical tools and assess performance under controlled wave conditions, primarily using the open-source software WECSim [11]. This solid foundation and open availability of performance data make the RM3 an ideal platform for conducting detailed parametric investigations into the influence of floater geometry on hydrodynamic performance and energy conversion in two-body systems.

This work conducts a comprehensive parametric investigation of the hydrodynamic behavior of two-body PA WECs, with a particular focus on the RM3 configuration. The study systematically evaluates how variations in floater geometry influence key performance metrics, including hydrodynamic coefficients, relative motion and power capture. By revealing new geometry-dependent relationships and insights, the analysis addresses the identified gap and supports more informed design decisions. The main contribution is the development of a quantitative parametric analysis of geometric sensitivity in two-body PA WECs, evaluating its impact on energy absorption and hydrodynamic system response under realistic sea states. An open-source code implementation is available at [GitHub](https://github.com/Energy-Control-and-Optimization-Lab/WEC-Tunable-RM/tree/IDETC2026)¹ to promote reproducibility, validation, and future extensions.

The paper is organized as follows. Section 2 introduces the problem formulation, modeling assumptions, and the architecture of the two-body PA system. Section 3 develops the numerical model governing the system dynamics and power capture under realistic sea states. Section 4 describes the parametric assessment methodology, and Section 5 presents the corresponding simulation results, which are further analyzed in Section 6 to examine the influence of geometry on hydrodynamic performance. Finally, Section 7 summarizes the key contributions and outlines directions for future research.

2. TWO-BODY POINT ABSORBER WEC: REFERENCE MODEL 3

This study focuses on a two-body PA WEC, drawing on the well-established RM3 [9], developed under the SANDIA National Laboratories (SNL) RMP. RM3 corresponds to a two-body PA configuration, where a floating buoy oscillates vertically relative to a submerged reaction structure in response to ocean wave excitation. The induced relative heave motion between the two bodies is harnessed to drive the Power Take-Off (PTO) system, converting mechanical energy into usable electrical power. The main design parameters for RM3 and their dimensions are presented in Fig. 1.

The RM3 was originally designed for the site located off the coast of Eureka, in Humboldt County, California. The sea states at this location exhibit significant wave heights (H_s) ranging from 0.5 [m] to 6 [m], and peak periods (T_p) from 5 [s] to 18 [s], over water depths between 40 [m] and 100 [m] [9]. The rated power output of a single WEC is 300 [kW], based on numerical simulations performed with WEC-Sim, assuming a capacity factor of 30%. Under these conditions, the estimated annual energy production per unit is approximately 700 [MWh] [9].

¹<https://github.com/Energy-Control-and-Optimization-Lab/WEC-Tunable-RM/tree/IDETC2026>

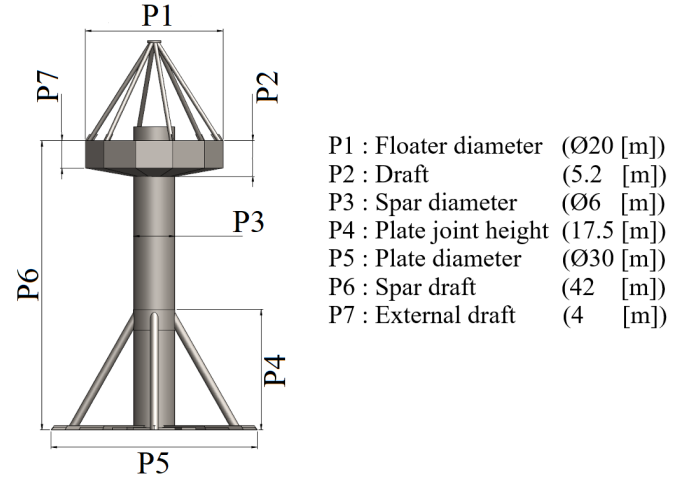


FIGURE 1: Schematic of Reference Model 3 (RM3) and its main dimensions [12].

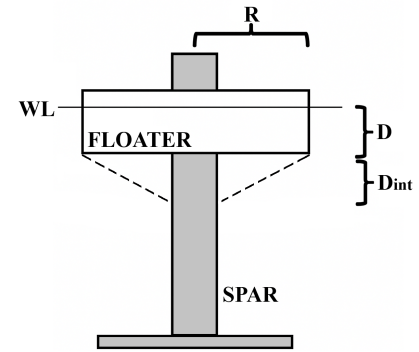


FIGURE 2: Simplified diagram of the two-body PA WEC under study.

In this paper, a parametric analysis of the ideal average power output is carried out for two representative sea states using a *simplified representation of the RM3 geometry*. The analysis is limited to the heave (vertical motion) Degree of Freedom (DOF), which constitutes the dominant motion mode in PA systems and the principal contributor to power absorption. The investigation considers systematic variations of the floater dimensions, specifically the radius (R) and external draft (D), while preserving a constant submerged volume, in addition to variations in the internal draft (D_{int}), which are illustrated in Fig. 2. This framework enables a structured assessment of the impact of geometric parameters on the device's power capture performance.

3. NUMERICAL MODEL

3.1. Governing equations

For the numerical analysis of the two-body PA, the open-source Boundary Element Method (BEM) solver CAPYTAIN is employed, which is based on potential theory [5]. The BEM codes can be applied for the analysis of flow around floating bodies under waves as it allows the calculation of resulting pressure fields and the exerted hydrodynamic forces on the bodies. In the code, due to the potential flow assumption, only Laplace and Bernoulli Equations [13] need to be solved (Eq. 1 and Eq. 2 respectively):

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0 \quad (1)$$

$$\rho \frac{\partial \Phi}{\partial t} + p + \rho g z = p_0 \quad (2)$$

Where Φ is velocity potential, ∇^2 is Laplace operator, ρ is fluid density, p is pressure, p_0 is atmospheric pressure and z is water depth.

3.2. Equations of motion

Equations 1 and 2 are solved for the 12 DOF of the system, comprising 6 DOF for the floater and 6 DOF for the submerged spar. For each body, there are three translational displacements (surge along the x-axis, sway along the y-axis, and heave along the z-axis) and three rotational motions (roll about the x-axis, pitch about the y-axis, and yaw about the z-axis).

The system of equations presented above is solved applying the kinematic and dynamic boundary conditions at the free surface, and impermeability boundary condition at the bottom and body surfaces [14]. Additionally, equations of motion written in terms of hydrodynamics coefficients in frequency domain [13] are presented in Eq. 3 in matrix form. In this equation, A is added mass matrix, M is body mass matrix, B damping matrix, C restoration matrix, F_e represents the external wave forces vector, and ω represent the angular frequency of the wave. Thus, the variable vector ξ stands for the 12 DOF of the system. Then, the Response Amplitude Operator (RAO), the solution in the frequency domain, is given by Eq. 4.

$$(A_{12 \times 12} + M_{12 \times 12}) \ddot{\xi}_{12 \times 1} + B_{12 \times 12} \dot{\xi}_{12 \times 1} + C_{12 \times 12} \xi_{12 \times 1} = F_{e12 \times 1} \quad (3)$$

$$\xi_{12 \times 1} = [- (M_{12 \times 12} + A_{12 \times 12}) \omega^2 + B_{12 \times 12} \omega i + C_{12 \times 12}]^{-1} F_{e12 \times 1} \quad (4)$$

Considering Eq. 4, it is possible to evaluate an additional parameter, namely the relative heave displacement between the two bodies, which can be expressed as $\xi_{rel} = \xi_3 - \xi_9$. Here, the subscripts 3 and 9 identify the heave motions of the floater and the spar, respectively. For a more comprehensive derivation and discussion of these equations, the reader is referred to [13].

3.3. Sea states

The JONSWAP (Joint North Sea Wave Project) spectrum constitutes a modification of the Pierson–Moskowitz spectrum, incorporating a peak enhancement factor to better capture the characteristics of developing sea states [15], as expressed in Eq. 5.

$$S_w(\omega) = \alpha g^2 \omega^{-5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp\left[-\frac{(\omega - \omega_p)^2}{2\sigma^2 \omega_p^2}\right]} \quad (5)$$

where $S(\omega)$ is the wave energy spectral density, ω is the wave angular frequency, ω_p is the angular peak frequency, α is the Phillips parameter (a scale factor), γ is the peak enhancement factor (typically $\gamma \approx 3.3$), g is the gravitational acceleration, and σ is the spectral width parameter, usually defined as $\sigma = 0.07$ for $\omega \leq \omega_p$ and $\sigma = 0.09$ for $\omega > \omega_p$.

The response of the device subjected to irregular wave spectrum is studied by the motion spectrum $S_\xi(\omega)$ [13], defined in Eq. 6:

$$S_\xi(\omega) = S_w(\omega) \cdot \xi^2 \quad (6)$$

The n -th order spectral moment of the motion process is obtained from the motion spectrum [13], defined in Eq. 7:

$$m_n = \int_0^\infty \omega^n S_\xi(\omega) d\omega. \quad (7)$$

These moments are used for the estimation of statistical properties of displacement in irregular wave conditions. An estimate of the significant displacement is given by [13]: $\xi_s \approx 4\sqrt{m_0}$.

3.4. Available power

The maximum absorbed power by a two-body PA considering the relative heave motion, is given by Eq. 8, where $\hat{F}_i(\omega)$ denotes the complex amplitude of the vertical excitation force, and $B_{eff}(\omega)$ is the effective radiation damping of the relative mode, with $B_{ij}(\omega)$ (Eq. 9) representing the radiation damping coefficients. This expression corresponds to the theoretical upper bound of absorbed power under optimal reactive control, when the PTO damping is matched to the effective radiation damping [16, 17].

$$P_{max}(\omega) = \frac{|\hat{F}_{e3}(\omega) - \hat{F}_{e9}(\omega)|^2}{8 B_{eff}(\omega)} \quad (8)$$

$$B_{eff}(\omega) = B_{3,3}(\omega) + B_{9,9}(\omega) - 2 B_{3,9}(\omega) \quad (9)$$

The average power absorbed by a WEC in a realistic sea state, assuming an ideal PTO that matches the effective radiation damping at each frequency, can be expressed as shown in Eq. 10 [15, 16]:

$$P_{avg} = \int_0^\infty P_{max}(\omega) S(\omega) d\omega \quad (10)$$

4. PARAMETRIC HYDRODYNAMIC ANALYSIS METHODOLOGY

4.1. Assumptions

The hydrodynamic analysis of the two-body PA is performed using linear potential flow theory. Only radiation damping (B), added mass (A), and hydrostatic restoring forces (F_e) are considered, as obtained from the BEM solution. Viscous damping (B_v) is neglected in order to focus on the theoretical upper bound of the absorbed power. This assumption is common in frequency-domain studies, as the fluid is considered inviscid and irrotational, and provides a consistent reference for comparison [18]. In addition, operational limits on the body motions are considered, specifically, the relative heave motion between the two bodies is constrained not to exceed 4 [m] under the evaluated sea specters, according to the design of RM3 [9].

4.2. Parametrization

The reference dimensions of our investigated device are based on RM3 model. To facilitate the parametric study, the device components are simplified into basic geometric shapes, as shown in Fig. 3. The design space for the parametric assessment focuses on the floater, defined by the key design variables: radius

(R) and external draft (D), which are systematically varied, while the spar geometry is kept fixed. These variations maintain the same total mass and submerged volume for each floater, ensuring consistent displacement characteristics across all configurations. The relation between these parameters are given in Eq. 11:

$$D = \frac{273}{R^2 - 9} \quad (11)$$

The hydrodynamic coefficients (A , B , and F_e) are calculated using the CAPYTAIN BEM solver. These coefficients are subsequently employed to predict the device response under realistic sea states characterized by a JONSWAP spectrum, enabling the definition of operational design limits based on *maximum vertical displacement*. Configurations that fail to satisfy these criteria are discarded, and the mean power output is evaluated for the remaining designs. Finally, an additional variation is introduced by modifying the internal draft (D_{int}), thereby altering the submerged volume and generating a geometric angle in the floater. This variation is analyzed to assess its influence on the expected mean ideal power output, providing further insight into the impact of the floater geometry on device performance. Fig. 3 illustrates the floater variations for different values of R , D , and D_{int} .

4.3. Convergence analysis

A mesh convergence analysis is carried out to determine the appropriate discretizations of the floater geometry. The objective is to ensure that the number of panels used is sufficient to provide accurate and stable results, while avoiding unnecessary computational cost. The convergence study is performed by progressively increasing the number of panels in the mesh (N_{pan}) and evaluating the corresponding hydrodynamic coefficients (A , B , and F_e). The selected mesh density corresponds to the level at which further refinement led to negligible variations in these coefficients, thereby guaranteeing both reliability and computational efficiency in the subsequent simulations. Based on this analysis, a total of 3200 panels is chosen for the floater discretizations and 5122 for the spar.

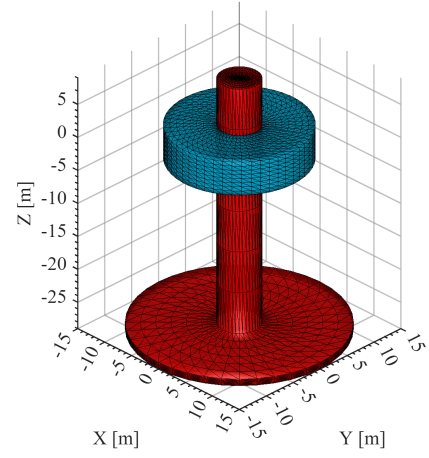
5. SIMULATION RESULTS

5.1. Hydrodynamical coefficients

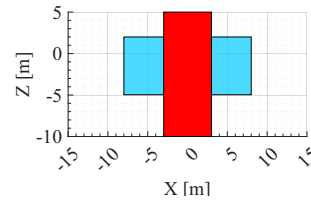
The hydrodynamic coefficients A , B , and F_e for the heave DOF of both bodies are shown in Fig. 4. For the spar, these coefficients remain constant, whereas for the floaters they vary with the ratio R/D (with $D_{int} = 0$). As R/D increases, A , B , and F_e for the floaters all rise. The spar has consistently higher A but lower B and F_e than most floaters, with the gap widening at larger R/D . This shows how floater geometry, under constant submerged volume, affects hydrodynamic coefficients and energy extraction potential.

5.2. Motion response

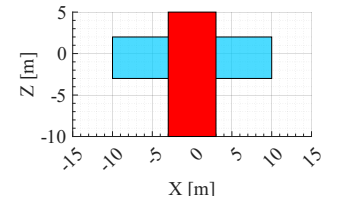
For the motion response analysis under real sea states, data from the National Data Buoy Center (NDBC) at station 46022 off Eureka, California [19] is used. Two representative conditions are defined by the mean Peak Period (T_p , taken as the dominant period) and mean Significant Wave Height (H_s) for August (Sea State 1) and December (Sea State 2) from 1982–2008 [19], representing the lowest and highest H_s , respectively.



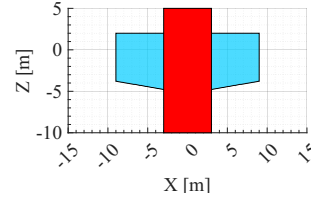
(a) Panoramic view of the virtual model of the floater (blue) and the spar (red) for CAPYTAIN.



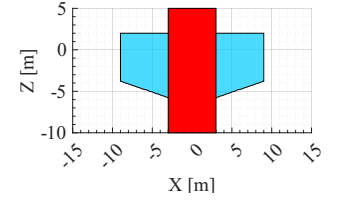
(b) Floater type 1:
 $R = 8$ [m],
 $R/D = 1.61$,
 $D_{int} = 0$ [m]



(c) Floater type 2:
 $R = 10$ [m],
 $R/D = 3.33$,
 $D_{int} = 0$ [m]



(d) Floater type 3:
 $R = 9$ [m],
 $R/D = 2.37$,
 $D_{int} = 1$ [m]



(e) Floater type 4:
 $R = 9$ [m],
 $R/D = 2.37$,
 $D_{int} = 2$ [m]

FIGURE 3: Floater variations corresponding to different R/D (radius to the external draft ratio) and D_{int} (internal draft). The baseline geometry and dimensions of RM3 are represented in (c).

Both states were modeled with a JONSWAP spectrum, with spectral responses shown in Fig. 5(a). Sea State 1 has $H_s = 1.7$ [m] and $T_p = 8.9$ [s], while Sea State 2 is more energetic with $H_s = 3.3$ [m] and $T_p = 13.1$ [s].

Based on the calculated coefficients presented in Fig. 4, the relative heave RAO is obtained according to Eq. 4, with the corresponding results illustrated in Fig. 5(b) for each floater configuration defined by the R/D ratio with $D_{int} = 0$. The relative heave motion spectrum was then computed using Eq. 6 for Sea States 1 and 2, with the results shown in Fig. 5(c) and (d), respectively. For each condition, the significant relative heave displacement (ξ_s) was estimated according to $\xi_s \approx 4\sqrt{m_0}$, discarding the designs in which the operational limit (4 [m]) was exceeded. The results of

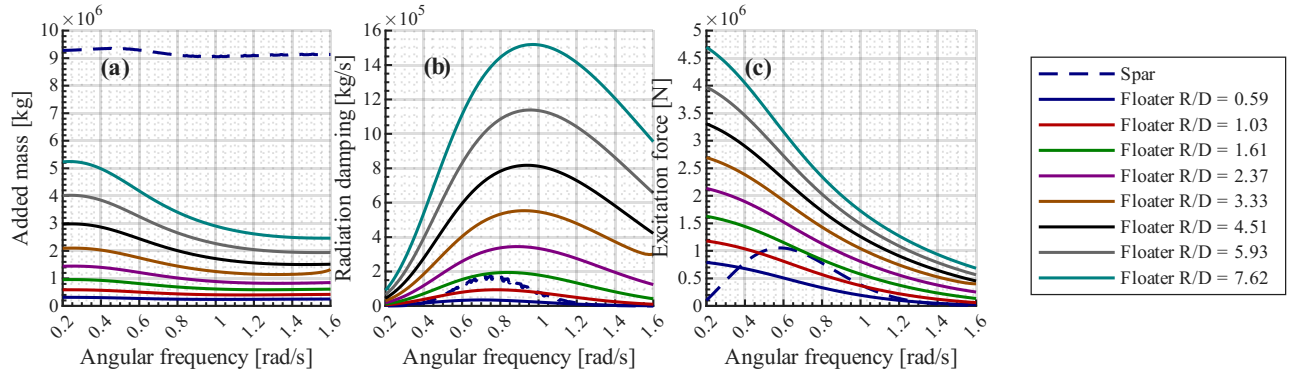


FIGURE 4: Hydrodynamical coefficients for different relations R/D , where (a) is the added mass (A), (b) the radiation damping (B) and (c) the excitation force (F_e).

TABLE 1: Significant relative heave displacement on heave motion (ξ_s), where R/D 1.03 and 0.59 exceed the operational limit of 4[m], as highlighted in red.

R/D	Sea state 1 (ξ_s [m])	Sea state 2 (ξ_s [m])
0.59	6.74	7.08
1.03	4.00	4.73
1.61	3.06	3.83
2.37	2.61	3.44
3.33	2.36	3.23
4.51	2.20	3.10
5.93	2.08	3.01
7.62	1.98	2.93

these relative displacements are shown in Table 1, highlighting the values that exceed the operational limit in red.

5.3. Power

To estimate the ideal maximum power as a function of wave frequency, Eq. 8 and Eq. 9 were applied. Subsequently, for each simulated sea state, the average power was evaluated for each geometry according to Eq. 10, with the results presented in Fig. 6, where the comparisons are shown with respect to the R/D ratio and the variations of D_{int} . For the evaluated sea states, a clear trend is observed in which the average ideal power increases as the R/D ratio decreases, while variations in D_{int} do not show a clear tendency nor a significant impact on the average power for either case.

6. DISCUSSION

This study analyzes the impact of floater geometry on system dynamics and energy capture under realistic sea conditions. The main design variables considered are the radius-to-external draft ratio (R/D) and the internal draft (D_{int}). Understanding how these parameters influence hydrodynamics is critical for guiding early-stage design and optimizing performance in the context of control co-design studies. The analysis focuses on heave motion, as it dominates energy extraction for this system. While spar geometry variations had little effect on hydrodynamic coefficients, the floater geometry showed a clear trend: as R/D increases, coefficients A, B, and F_e increase, while the relative heave RAOs

indicate that smaller R/D ratios amplify heave motion and shift the peak period to shorter values, highlighting the strong interplay between geometry and dynamic response.

The study also evaluates device performance under two representative sea states based on NDBC records for the RM3 deployment site. Sea state 1 has lower energy density with waves near 0.5 rad/s, while Sea state 2 features higher energy and longer periods below 0.6 [rad/s]. The resulting relative heave motion spectra show a dominant peak corresponding to the main resonance and a smaller secondary peak nearly constant across geometries. Lower R/D ratios increase the dominant peak, demonstrating the importance of aligning device natural frequency with the wave spectrum to maximize energy capture. Operational constraints, particularly a 4 m vertical displacement limit between spar and floater, led to the exclusion of geometries with R/D values of 0.59 and 1.03, emphasizing the balance between energy performance and structural reliability.

Finally, the effect of geometric parameters on mean power capture was quantified. The R/D ratio has a large influence, with an average normalized sensitivity of -20% , whereas D_{int} has only a marginal effect. Lower R/D values reduce the relative heave natural frequency, better aligning with dominant wave frequencies, enhancing resonance, amplifying RAO peaks, broadening the motion spectrum, and increasing energy capture potential. This analysis highlights R/D as the key geometric parameter for optimizing WEC performance under varying sea conditions.

7. CONCLUSIONS AND FUTURE WORKS

This study conducted a parametric analysis of floater geometries for a two-body PA, based on the RM3 model, highlighting the dominant role of the R/D ratio on hydrodynamic response and energy capture, while D_{int} showed minor influence. Lower R/D values amplify RAO peaks and align natural resonance with sea state frequencies, enhancing power but increasing operational risks. These insights emphasize the importance of early-stage geometry selection and provide a framework for integrating geometry-dependent effects into the design process. The next stage of this research will expand the analysis using a Design of Experiments approach to systematically identify optimal geometrical dimensions and the most influential parameters affecting energy extraction efficiency. This extended study will also incorporate a linear PTO system, modeled as an equivalent damping

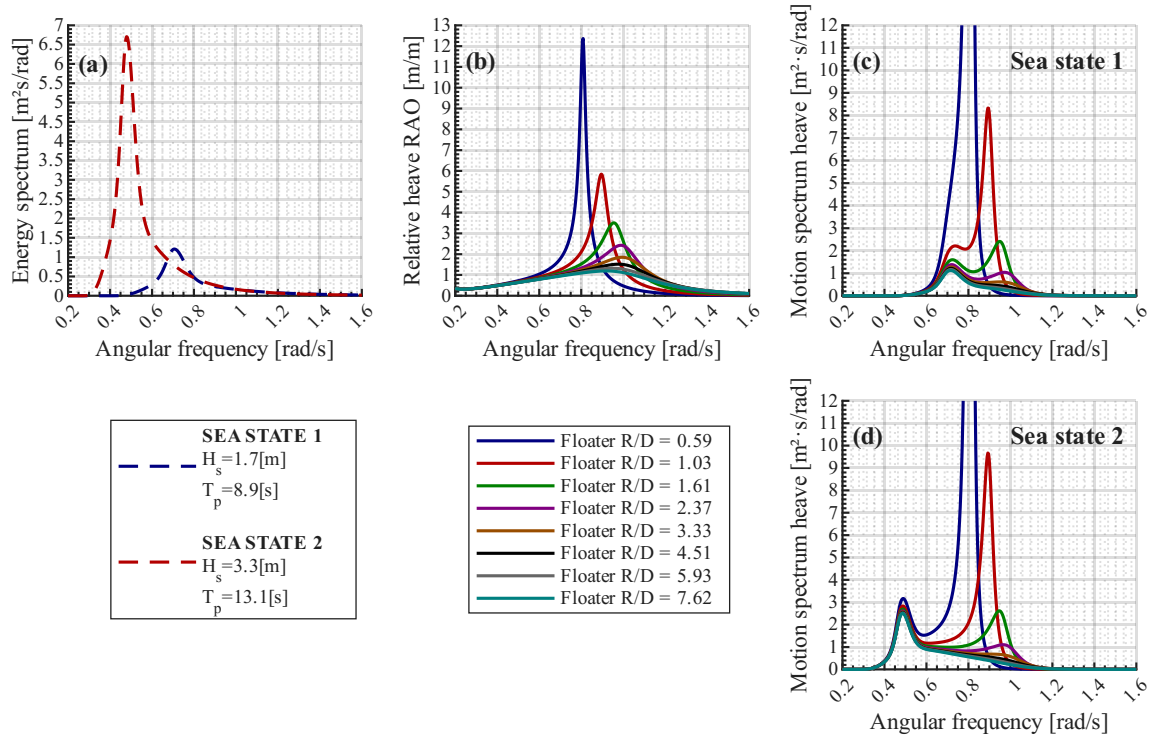


FIGURE 5: Response Amplitude Operator (RAO) for heave motion (b) and motion spectrum (c and d) according irregular sea state spectrums (a).

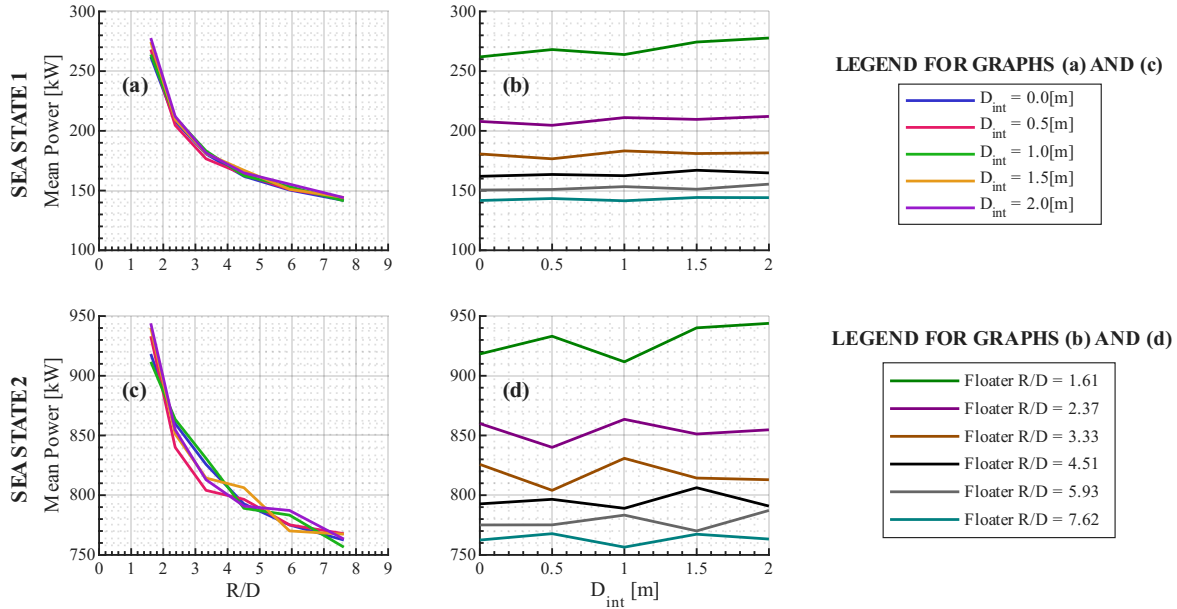


FIGURE 6: Mean maximum power capture of each WEC according R/D ratio in panels (a) and (c), while D_{int} variation is represented in panels (b) and (d). The first row is for Sea state 1, while the second is for Sea state 2.

coefficient, to more realistically capture the energy conversion dynamics under operational constraints.

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